

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)****Y25 - 2026-2027 Trimester - 4****Project Abstract Submission Form**

Sec No - 3	Team No - 07	Project Title - Ticket Booking System
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1. Abstract

The Ticket Booking System is a web-based application developed to simplify and automate the process of booking tickets for movies, events, or other entertainment services. Traditional ticket booking methods may require users to visit booking counters or use multiple platforms to check availability and reserve tickets. This system provides a single online platform where users can register, log in, browse available shows, select their preferred date and time, view the available seats, choose seats, and complete the booking process conveniently. After successful payment, the system generates a unique booking ID and digital ticket containing important booking details.

The system is designed using modern technologies such as React for developing an interactive and user-friendly frontend, Java and Spring Boot for implementing the backend services, and MySQL for storing user, movie/event, seat, booking, and payment information. The application provides different modules for users and administrators. Users can search for shows, select seats, make payments, view booking history, and cancel tickets when required. Administrators can add and manage shows, update seat availability, monitor bookings, manage users, and generate reports. This helps reduce manual work and provides faster access to booking information.

One of the major challenges addressed by the system is concurrent booking, where multiple users may attempt to book the same seat at the same time. To solve this problem, Redis distributed locking is used to temporarily lock a selected seat during the booking process and prevent multiple users from reserving the same seat. RabbitMQ is used as a message queue to process booking and payment requests efficiently, which helps the system handle a large number of users and requests. The system also supports cancellation and refund processing, making it more flexible and reliable. Overall, the proposed Ticket Booking System provides a secure, scalable, reliable, and concurrency-safe solution that improves the user experience while preventing double booking and seat overselling.

2. Team Details

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3. Frameworks / Platforms / Software Used

- **Java** – Used to develop the backend logic.
- **Spring Boot** – Used to create REST APIs and backend services.
- **React.js** – Used to design the user interface.
- **MySQL** – Used to store users, events, seats, bookings, and payments.
- **Redis** – Used for distributed locking to prevent two users from booking the same seat.
- **RabbitMQ** – Used to handle multiple booking requests through a queue.
- **Maven** – Used to manage Spring Boot dependencies and build the Java project.
- **Visual Studio Code** – Used as the development environment.
- **Postman** – Used to test backend APIs.

Guide Name and Signature (remarks if any) _____